ML Interview Notes

Machine Learning Fundamentals — Interview Preparation

Q1. How would you tackle multicollinearity in multiple linear regression?

Multiple linear regression is a method that uses several independent variables to predict or explain the dependent variable of interest. When using this technique, we assume that the independent (explanatory) variables are also independent from one another — i.e., their values do not affect one another.

Multicollinearity occurs when different independent variables are correlated. If the correlation between variables is high enough, it can cause problems in fitting the linear regression model.

Ways to tackle multicollinearity:

- **1. Determine if you actually care about this problem.**

  If you are solely interested in having a good predictor and are satisfied that your model does well on both the training and test sets, or if the correlation between variables is below a certain threshold, you may be able to leave this problem alone.

- **2. Reduce the number of independent variables.**

  (a) Remove some or most of the correlated variables and re-run your analysis.

  (b) Use a dimensionality reduction technique, such as PCA or a clustering approach, to lower the number of variables.

- **3. Standardize your independent variables.**

Q2. How would you interpret coefficients of logistic regression for categorical and boolean variables?

Boolean variables:

- The **magnitude** of the coefficient is directly correlated to its effect on the outcome probability.
- The **sign** of the coefficient tells you whether the variable is directly or inversely correlated with the outcome probability.

Categorical variables:

One-hot encode categorical variables into boolean variables, then apply the guidelines above.

Q3. When to use Random Forest over SVM and vice versa?

- **Multiclass problems:** RF is intrinsically suited for multiclass problems, while SVM is intrinsically a two-class classifier. For multiclass problems with SVM, you must reduce it into multiple binary classification problems.

- **Feature types & scaling:** RF works well with a mixture of numerical and categorical features and is robust to features on various scales — you can generally use data as-is. SVM maximises the margin and relies on the concept of distance between points, so one-hot encoding for categorical features is a must, and min-max (or other) scaling is highly recommended as a preprocessing step.

- **Scalability:** With n data points and m features, SVM constructs an $n \times n$ matrix as an intermediate step, making it hardly scalable beyond 10^5 points. A large number of features is generally not a problem for SVM.

- **Output type:** For a classification problem, RF gives you the probability of belonging to a class. SVM gives you the distance to the decision boundary; you still need to convert this to a probability (e.g., via Platt scaling).

- **Performance:** For problems where SVM is applicable, it generally performs better than Random Forest.

- **Interpretability:** SVM gives you "support vectors" — the points in each class closest to the decision boundary. These may be of interest in themselves for interpretation.

Q4. When to use Lasso, Ridge, and Elastic Net?

- **Lasso & Elastic Net** tend to give sparse weights (most zeros), because L1 regularization drives weights — both large and small — towards zero equally. If you have many predictors and suspect that not all of them are important, Lasso or Elastic Net may be a good starting point.

- **Ridge** tends to give small but well-distributed weights, because L2 regularization penalises large weights more than small ones — it shrinks big weights towards zero rather than driving any weight exactly to zero. If you only have a few predictors and are confident that all of them are relevant, try Ridge as a regularised linear regression model.

- **Scaling:** You must scale your data before using any of these regularised linear regression models.

Q5. What is the relation between the log-likelihood loss function for logistic regression and maximum likelihood estimation?

To choose values for the parameters of logistic regression, we use **Maximum Likelihood Estimation (MLE)**. The log-likelihood equation is:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{m} \Big[y^{(i)} \log \sigma\big(\boldsymbol{\theta}^\top \mathbf{x}^{(i)}\big) + \big(1 - y^{(i)}\big) \log\big(1 - \sigma\big(\boldsymbol{\theta}^\top \mathbf{x}^{(i)}\big)\big) \Big]$$

MLE chooses the parameters $\boldsymbol{\theta}$ that **maximise** $\mathcal{L}(\boldsymbol{\theta})$. Equivalently, gradient descent minimises the negative log-likelihood $-\mathcal{L}(\boldsymbol{\theta})$. The parameter update rule is:

$$\theta_j^{(\text{new})} = \theta_j^{(\text{old})} - \eta \cdot \frac{\partial \mathcal{L}(\boldsymbol{\theta}^{(\text{old})})}{\partial \theta_j^{(\text{old})}} = \theta_j^{(\text{old})} - \eta \sum_{i=1}^{m} \Big[y^{(i)} - \sigma(\boldsymbol{\theta}^\top \mathbf{x}^{(i)}) \Big] x_j^{(i)}$$

Q6. What is the loss function for SVM?

Linear SVM classifier prediction:

$$\hat{y} = \begin{cases} 0 & \text{if } \mathbf{w}^\top \mathbf{x} + b < 0 \\ 1 & \text{if } \mathbf{w}^\top \mathbf{x} + b \geq 0 \end{cases}$$

Hard-margin linear SVM classifier objective:

$$\underset{\mathbf{w},\, b}{\text{minimise}} \quad \tfrac{1}{2}\, \mathbf{w}^\top \mathbf{w} \qquad \text{subject to} \quad t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) \geq 1 \quad \forall\, i$$

where $t^{(i)} = -1$ for negative instances ($y^{(i)} = 0$) and $t^{(i)} = +1$ for positive instances ($y^{(i)} = 1$).

Soft-margin linear SVM classifier objective:

$$\underset{\mathbf{w},\, b,\, \boldsymbol{\xi}}{\text{minimise}} \quad \tfrac{1}{2}\, \mathbf{w}^\top \mathbf{w} + C \sum_{i=1}^{m} \xi^{(i)} \qquad \text{subject to} \quad t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b) \geq 1 - \xi^{(i)}, \quad \xi^{(i)} \geq 0 \quad \forall\, i$$

Equivalently (substituting out $\xi^{(i)}$):

$$\underset{\mathbf{w},\, b}{\text{minimise}} \quad \tfrac{1}{2}\, \mathbf{w}^\top \mathbf{w} + C \sum_{i=1}^{m} \underbrace{\max(0,\ 1 - t^{(i)}(\mathbf{w}^\top \mathbf{x}^{(i)} + b))}_{\text{hinge loss}}$$

$\xi^{(i)}$ measures how much the i-th instance is allowed to violate the margin.

Q7. When to use logistic regression vs. SVMs?

Let n = number of features ($\mathbf{x} \in \mathbb{R}^{n+1}$) and m = number of training examples.

Scenario	Example	Recommendation
n **large relative to** m	$n \geq m$; e.g. $n = 10{,}000$, $m = 10$–$1{,}000$	Logistic regression, or SVM with linear kernel
n **small,** m **intermediate**	$n = 1$–$1{,}000$, $m = 10$–$10{,}000$	SVM with Gaussian (RBF) kernel
n **small,** m **large**	$n = 1$–$1{,}000$, $m = 50{,}000+$	Create/add more features, then use logistic regression or SVM with linear kernel